In [1]:
```
import numpy as np
import pandas as pd
```

In [2]:
```
df = pd.read_csv('customer_shopping_behavior.csv')
```

In [3]:
```
df.head()
```

Out[3]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Disc Ap
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	

In [4]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

In [5]:
```
# summary stats
df.describe(include='all')
```

Out[5]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN

In [6]:
```
# check null values
df.isnull().sum()
```

Out[6]:
```
Customer ID                0
Age                        0
Gender                     0
Item Purchased             0
Category                   0
Purchase Amount (USD)      0
Location                   0
Size                       0
Color                      0
Season                     0
Review Rating             37
Subscription Status        0
Shipping Type              0
Discount Applied           0
Promo Code Used            0
Previous Purchases         0
Payment Method             0
Frequency of Purchases     0
dtype: int64
```

In [7]:
```
# impute null vslues in review by imputing median of product category

df['Review Rating'] = df.groupby('Category')['Review Rating'].transform( lambda x: x.fillna(x.median()))
```

In [8]:
```
df.isnull().sum()
```

Out[8]:
```
Customer ID               0
Age                       0
Gender                    0
Item Purchased            0
Category                  0
Purchase Amount (USD)     0
Location                  0
Size                      0
Color                     0
Season                    0
Review Rating             0
Subscription Status       0
Shipping Type             0
Discount Applied          0
Promo Code Used           0
Previous Purchases        0
Payment Method            0
Frequency of Purchases    0
dtype: int64
```

In [9]:
```
# renaming columns to snake case
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ','_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

In [10]:
```
df.columns
```

Out[10]:
```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
       'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

In [11]:
```
# feature engineering
# create new age group column

labels = ['young adult','adult','middle-aged','senior']
df['age_group'] = pd.qcut(df['age'], q = 4, labels = labels )
# qcut divides into q=4 equal parts
```

In [12]:
```
df[['age','age_group']].head()
```

Out[12]:

	age	age_group
0	55	middle-aged
1	19	young adult
2	50	middle-aged
3	21	young adult
4	45	middle-aged

In [13]:
```
# new columhn: purchase frequency days

freq_map = {
    'Fortnightly': 14,
    'Weekly':7,
```

```
    'Monthly': 30,
    'Quarterly':90,
    'Bi-Weekly': 14,
    'Annually': 365,
    'Every 3 Months': 90
}

df['purchase_frequency_days'] = df['frequency_of_purchases'].map(freq_map)
```

In [14]:
```
df[['purchase_frequency_days','frequency_of_purchases']].head(10)
```

Out[14]:

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly

In [15]:
```
# cehck for redundant columns
df['discount_applied'] == df['promo_code_used'].all()
```

Out[15]:
```
0       False
1       False
2       False
3       False
4       False
        ...
3895    False
3896    False
3897    False
3898    False
3899    False
Name: discount_applied, Length: 3900, dtype: bool
```

In [16]:
```
# delete promo code used
df = df.drop('promo_code_used',axis=1)
df.columns
```

Out[16]:
```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'previous_purchases', 'payment_method',
       'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],
      dtype='object')
```

connecting python script to mysql

In [17]:
```
!pip install pymysql sqlalchemy
```

```
Requirement already satisfied: pymysql in c:\users\lenova\downloads\newana\envs\dsfs\lib\site-packages (1.1.2)
Requirement already satisfied: sqlalchemy in c:\users\lenova\downloads\newana\envs\dsfs\lib\site-packages (2.0.48)
Requirement already satisfied: greenlet>=1 in c:\users\lenova\downloads\newana\envs\dsfs\lib\site-packages (from sqlalchemy)
(3.3.2)
Requirement already satisfied: typing-extensions>=4.6.0 in c:\users\lenova\downloads\newana\envs\dsfs\lib\site-packages (from s
qlalchemy) (4.12.2)
```

In [18]:
```
pip install psycopg2-binary sqlalchemy
```

```
Requirement already satisfied: psycopg2-binary in c:\users\lenova\downloads\newana\envs\dsfs\lib\site-packages (2.9.11)
Requirement already satisfied: sqlalchemy in c:\users\lenova\downloads\newana\envs\dsfs\lib\site-packages (2.0.48)
Requirement already satisfied: greenlet>=1 in c:\users\lenova\downloads\newana\envs\dsfs\lib\site-packages (from sqlalchemy)
(3.3.2)
Requirement already satisfied: typing-extensions>=4.6.0 in c:\users\lenova\downloads\newana\envs\dsfs\lib\site-packages (from s
qlalchemy) (4.12.2)
Note: you may need to restart the kernel to use updated packages.
```

In [19]:
```
from sqlalchemy import create_engine
from sqlalchemy.engine import URL

# mysql connection
url = URL.create(
    drivername = 'mysql+pymysql',
    username = 'root',
```

```
    password = 'bhagurkar123!@#',
    host = '127.0.0.1',
    port = '3306',
    database = 'customer_behavior' )

engine = create_engine(url)

# write datafrmae to mysql

table_name = 'customer'
df.to_sql(table_name, engine, if_exists='replace', index=False)

# read back sample
pd.read_sql('SELECT * FROM customer LIMIT 5;', engine)
```

Out[19]:

	customer_id	age	gender	item_purchased	category	purchase_amount	location	size	color	season	review_rating	subscrip
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	

In [ ]:

In [ ]: